Suphanat Chanlek

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PROFESSIONAL SUMMARY

Third-year Computer Science student with hands-on experience in full-stack, frontend, and backend development, along with exposure to AI/MLOps engineering, gained through internships and hackathons. Skilled in building and deploying production-ready web applications using modern frameworks, with a solid foundation in API design, database integration, Docker, and Git workflows. Seeking a Software Engineering Internship / Co-op (Full-Stack, Frontend, or Backend) to apply my development skills in designing responsive interfaces, implementing robust backend services, and contributing to scalable, maintainable software solutions.

EDUCATION

Thammasat University **Rangsit Campus, Thailand**

Bachelor of Science in Computer Science, 3.31 GPA 2024 - Present

- **Relevant Coursework:** Web Development, Object-Oriented Programming, Data Structures, Design and Analysis of Algorithms, Database Systems, Software Engineering, Mobile Application Development, Net-Centric Computing, Operating Systems.

TECHNICAL SKILLS

- **Backend:** Go, Java, Spring Boot, Node.js, Express.js, Nest.js, FastAPI, Kotlin, C#, .Net.
- **Frontend:** JavaScript, TypeScript, React.js, Next.js, Vue.js, Tailwind CSS, Bootstrap.
- **Database:** PostgreSQL, MySQL, MongoDB, Microsoft SQL Server, Redis, SQLite.
- **DevOps & Cloud:** Docker, Kubernetes, Docker Compose, Git, GitHub Actions, CI/CD Pipelines, Jenkins, Linux, Bash, Nginx, Terraform, AWS, GCP, Azure, Monitoring & Logging, Grafana.
- **ML/AI:** Python, NumPy, Pandas, Open CV, Scikit-learn, XGBoost, TensorFlow, PyTorch, Matplotlib, Seaborn.
- **Tools & Technologies:** Git, GitHub, GitLab, Postman, Figma, Firebase, Prisma, Label Studio, Supabase, K6.

WORK EXPERIENCE

T.C.C. Technology CO., LTD. **Bangkok**

Application Developer (Internship & Part-Time) June 2026 – Present

- Developed backend APIs and business logic for **SOOK, a (food delivery platform)** on microservices architecture.
- Fixed **IDOR vulnerabilities** across multiple services, closing security gaps that exposed other users' data.
- Diagnosed slow queries in production and proposed source-code-level optimizations.
- Fixed memory leaks and added APM tracing to improve stability and monitoring.
- Designed a safe migration plan to refactor a legacy coupon/promotion system.

CONNECTED TECH CO., LTD. **Bangkok**

AI / MLOps Engineer (Internship) July 2026 – Sep 2026

- Built AEGIS, an edge PPE-compliance detection system (Mac mini M4) reaching **0.962 macro-F1** at **23.6 FPS.**
- Designed the vision pipeline: pose estimation → tracking → ROI extraction → classification → decision fusion.
- Benchmarked 4 backbones on a self-curated 23,622-crop dataset; selected EfficientNet-B0 for fewest errors.
- Closed the real-world accuracy gap via calibration and hard-negative mining, boosting glove F1 from **0.70** → **0.86.**
- Shipped MQTT integration with outbox/retry resilience for reliable edge-to-cloud delivery.
- Built visual debug tooling that surfaced real-world failure modes offline metrics had missed.

Thammasat University – Lab Tonkit **Rangsit Campus**

Undergraduate Researcher (AI & Software Engineering) May 2025 – Present

- Contributed to the PADIN Project, a joint AI research collaboration between **Osaka University** and Thammasat University.
- Developed AI-powered dental X-ray diagnostic tools using SAM 2 and image processing techniques.
- Built an ML backend integrated with Label Studio to support dental image annotation and federated learning research.
- Enabled researchers to process approximately **2,000 dental X-ray images per month.**
- Developed and maintained the PADIN project website for research data and annotation tool access
- Deployed a web application on Render to combine individual tooth labels into composite dental images.

ASCEND CO., LTD. **Bangkok**

Frontend Developer (Internship) August 2025 – November 2025

- Developed and deployed responsive web interfaces using Next.js and Tailwind CSS for production systems.
- Integrated frontend components with various backend services, ensuring seamless real-world application deployments.
- Collaborated closely with design teams to translate Figma mockups into high-fidelity, production-ready code.
- Debugged and resolved UI issues reported in production, improving overall user experience.

TRAINING & PROFESSIONAL DEVELOPMENT

Super AI Engineer Season 6 (Silver Medal) Mar 2026 – Sep 2026

Artificial Intelligence Association of Thailand (AIAT)

- Selected as 1 of 156 participants from 10,457+ applicants nationwide **(top 1.5%)** for a 3-level national AI training program.
- Cleared Level 1–3 (AI Fundamentals → Intensive AI Camp → AI Internship), covering AI, ML, Computer Vision, NLP, LLMs, and RAG through theoretical assessments and hands-on projects.
- Solved real-world industry problems during the Intensive AI Camp under mentorship from AI professionals; awarded **Silver Medal** upon program completion.
- Ranked among top performers in Kaggle Mini-Hackathons, clearing baseline criteria across multiple AI tasks.

PROJECTS

Chulalongkorn University Building Information Catalog | Next.js, Go, PostgreSQL, AWS EC2, AWS Cognito, AWS S3

- Built and deployed a **production** BIM/IFC platform centralizing Chulalongkorn University building data.
- Developed an end-to-end BIM/IFC workflow for access, processing, tracking, and 3D visualization.
- Implemented Authentication, Authorization, and RBAC for secure BIM access, approval workflows, and usage tracking.

SOS Backoffice | Next.js, Nest.js, TypeScript, Tailwind CSS, PostgreSQL

- Built and owned the production backoffice frontend for a food-surplus platform serving **100+ stores** and communities.
- Built responsive, reusable UI components from Figma using Next.js, TypeScript, and Tailwind CSS.
- Implemented core flows (data tables, forms, filters, pagination) with robust state and error handling for production use.
- Integrated backend APIs and collaborated on API contracts and production issue resolution.

Real-Time ML Recommender System | Polars, LightGBM, Kafka, Redis, FastAPI, MLflow, Docker, Prometheus, Grafana

- Built a two-stage recommender (ALS + LightGBM LambdaRank) on 2.7M events, reaching NDCG@10 of 0.033
- Deployed real-time serving with Kafka, Redis and FastAPI, with MLflow versioning and Grafana monitoring.
- Load-tested with k6: p95 latency of 30 ms, about 1,360 req/s capacity, 0% errors.

Thai Gov Photo & Doc Processor | Go, Next.js, Terraform, Kubernetes (K3s), Jenkins, Argo CD, Trivy, AWS

- Built an end-to-end CI/CD pipeline from Terraform-provisioned AWS infrastructure (EC2, ECR, S3, least-privilege IAM) to GitOps deploys on K3s with Argo CD, using a photo converter for Thai government applications as the workload.
- Automated Jenkins CI on ephemeral Kubernetes agents: lint, test, build once, a Trivy gate that blocks CRITICAL CVEs, then push the scanned image to ECR tagged with the git SHA.
- Delivered zero-downtime rolling updates with a post-deploy smoke test and git revert rollback. Added nightly Terraform drift detection that alerts on Discord, and secured the cluster (non-root containers, IAM roles with no stored keys, SSM instead of SSH, HTTPS via Let's Encrypt).

HONORS & AWARDS

Hackathon:

The Winner, Sci Soft Power Innovation Hackathon (SSPI 2024)

- IN SCENE, a travel recommendation platform connecting Thai film locations with real-world destinations.
- Delivered an end-to-end prototype through cross-functional collaboration.

The Winner, Coffee Chain Hackathon 2026 - Demand Forecasting at Scale (Super AI Engineer Season 6 x ExpresSo NB)

- Built a hybrid demand forecasting system using CatBoost and Chronos-Bolt for inventory planning.
- Proposed AI-driven inventory optimization to reduce waste, prevent stockouts, and improve ordering decisions.

1st Runner-up, Sony Smart Factory Hackathon 2025 (SONY DEVICE TECHNOLOGY THAILAND)

- Built an AI-powered supply chain and production planning platform for factory operations.
- Developed smart procurement, anomaly detection, and risk analysis to support production planning.

2nd Runner-up, Edge-AI for Intelligence Transport System Hackathon 2026 (Super AI Engineer Season 6 x NT)

- Built a real-time Edge AI pipeline for multi-camera vehicle detection and classification.
- Reduced vehicle search time from 1 hour to < 1 minute, saving an estimated **฿451K/year**

Silver Medal, Super AI Engineer Season 6 (Artificial Intelligence Association of Thailand)

- Selected as 1 of 156 (top ~1.5%) from 10,457+ applicants nationwide; completed all 3 levels.
- Completed intensive AI/ML training covering Computer Vision, NLP, and LLMs, including an AI industry internship.

Rookie Star, Axtra Mile Hackathon 2025 (CP AXTRA)

- Developed AmaMove, an end-to-end mobility platform for underserved communities.
- Designed an AI-driven solution for personalized services and customer insights.

Selected participant, ICPC Thailand National Contest 2026 (Kasetsart University)

- advanced from Sub-Regional Round through algorithmic problem-solving under competition conditions.

Selected Participant (Top 20 out of 273 applicants), Young Webmaster Camp 20 – Web Programming Track

Selected Participant (Top 150 out of 10,457 participants), Super AI Engineer Season 6 – Level 2 Onsite Program